# Speech Processing Lab Assignment-6

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## 1 Title

**Pitch and Formant Estimation using FFT and STFT in Speech Processing with Spectrogram Analysis**

## 2 Objective

- To extract and visualize pitch, harmonics and formants using FFT
- To analyze speech using STFT and spectrograms
- To compare narrowband and wideband spectrograms
- To understand time-frequency representation of speech signals

## 3 Dataset

A speech sample was recorded and converted into `.wav` format. The audio signal was processed using Python libraries such as Librosa and NumPy.

## 4 Theory

### 4.1 FFT (Fast Fourier Transform)

FFT converts a time-domain signal into frequency-domain representation:

$$X(f) = \sum x(n)e^{-j2\pi fn}$$

It helps in identifying:

- Pitch (Fundamental Frequency)
- Harmonics
- Formants (approximate peaks)

## 4.2 STFT (Short Time Fourier Transform)

STFT analyzes signals over short windows:

$$X(t, f) = \sum x(n)w(n-t)e^{-j2\pi fn}$$

It provides a time-frequency representation.

## 4.3 Spectrogram

A spectrogram is a visual representation of STFT:

- X-axis → Time
- Y-axis → Frequency
- Color → Amplitude

## 4.4 Narrowband vs Wideband

- Narrowband → Better frequency resolution (harmonics visible)
- Wideband → Better time resolution (formants visible)

# 5 Implementation and Results

## 5.1 Audio Upload and Conversion

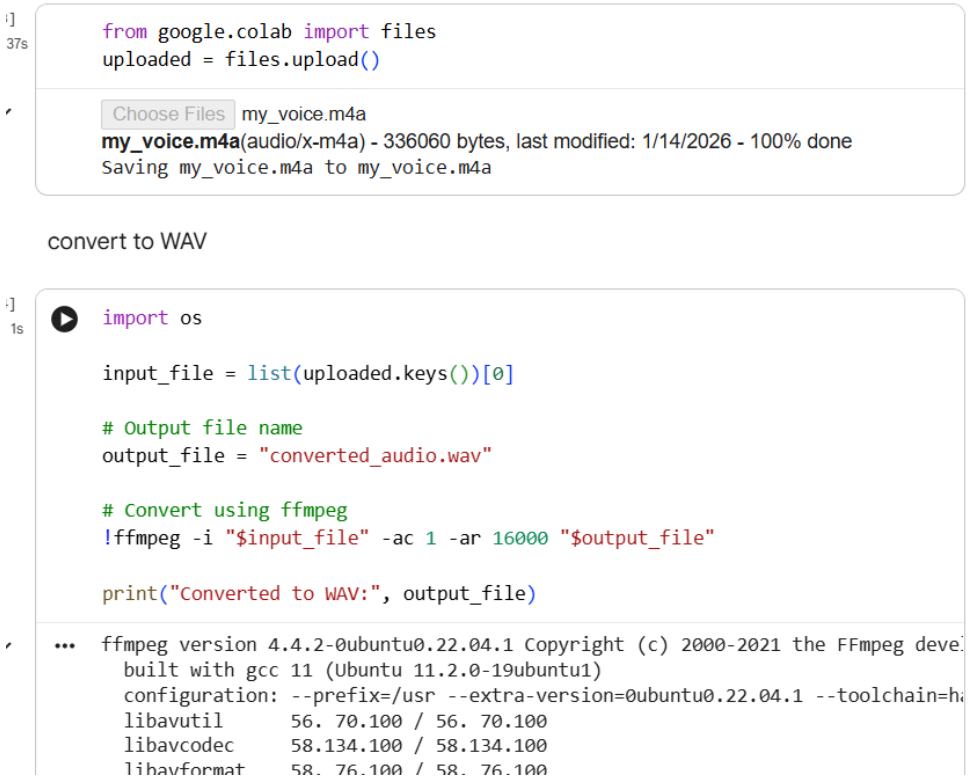


Figure 1: Audio Upload and Conversion to WAV

## 5.2 Speech Signal (Time Domain)

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
plt.plot(y)
plt.title("Speech Signal (Time Domain)")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.grid()
plt.show()
```

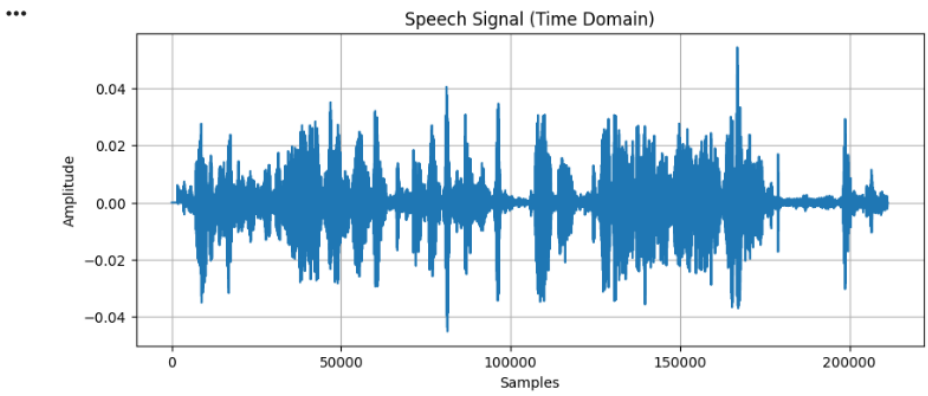


Figure 2: Speech Signal

## 5.3 FFT Spectrum

FFT ⟶ Pitch, Harmonics, Formants

```
import numpy as np

Y = np.fft.fft(y)
freqs = np.fft.fftfreq(len(Y), 1/sr)

plt.figure(figsize=(10,4))
plt.plot(freqs[:len(freqs)//2], np.abs(Y[:len(Y)//2]))
plt.title("FFT Spectrum")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

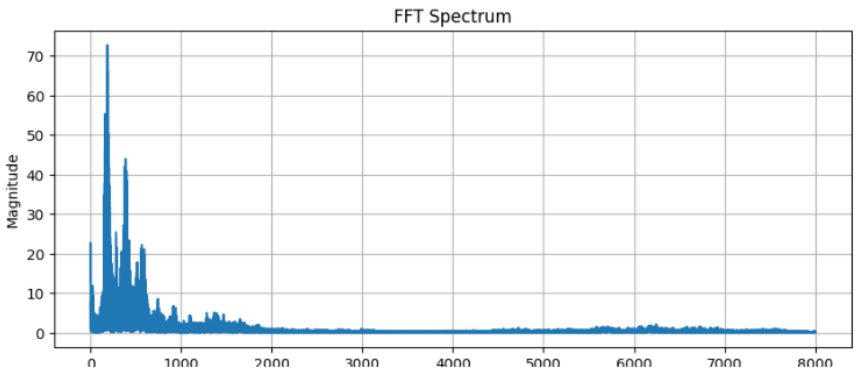


Figure 3: FFT Spectrum

**Observation:** The FFT spectrum shows dominant peaks corresponding to the fundamental frequency and harmonics.

## 5.4 Pitch Contour

```
f0, voiced_flag, voiced_probs = librosa.pyin(y, fmin=50, fmax=300)

plt.figure(figsize=(10,4))
plt.plot(f0)
plt.title("Pitch Contour (F0)")
plt.xlabel("Frames")
plt.ylabel("Frequency (Hz)")
plt.show()
```

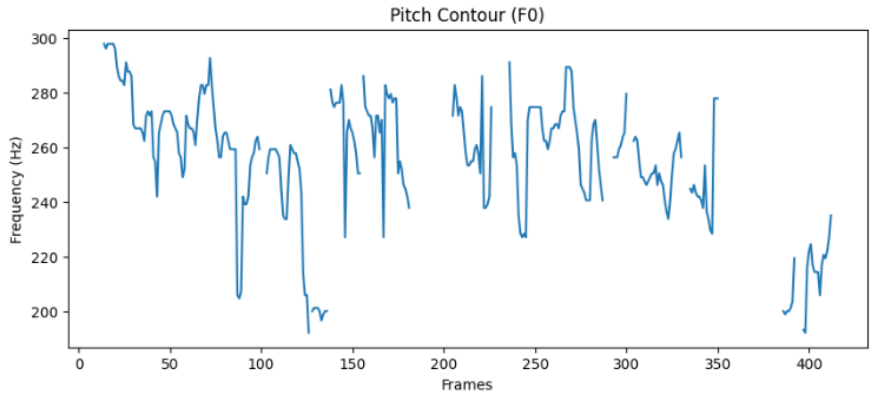


Figure 4: Pitch Contour (F0)

**Observation:** Pitch varies over time, indicating natural speech modulation.

## 5.5 Harmonics and Formants

```
spectrum = np.abs(Y[:len(Y)//2])
freqs_half = freqs[:len(freqs)//2]

peaks, _ = find_peaks(spectrum, height=max(spectrum)*0.1)

plt.figure(figsize=(10,4))
plt.plot(freqs_half, spectrum)
plt.plot(freqs_half[peaks], spectrum[peaks], "x")
plt.title("Harmonics / Formant Peaks")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()

print("Estimated Frequencies:", freqs_half[peaks[:5]])
```

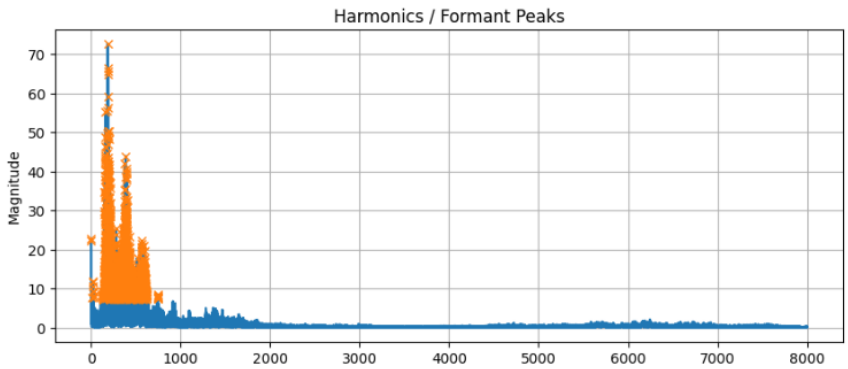


Figure 5: Harmonics and Formant Peaks

**Observation:** Multiple peaks represent harmonics, and some prominent peaks correspond to formant frequencies.

## 5.6 Spectrogram (STFT)

STFT → Spectrogram

```
D = librosa.stft(y)
S_db = librosa.amplitude_to_db(np.abs(D), ref=np.max)

plt.figure(figsize=(10,5))
librosa.display.specshow(S_db, sr=sr, x_axis='time', y_axis='hz')
plt.colorbar()
plt.title("Spectrogram (STFT)")
plt.show()
```

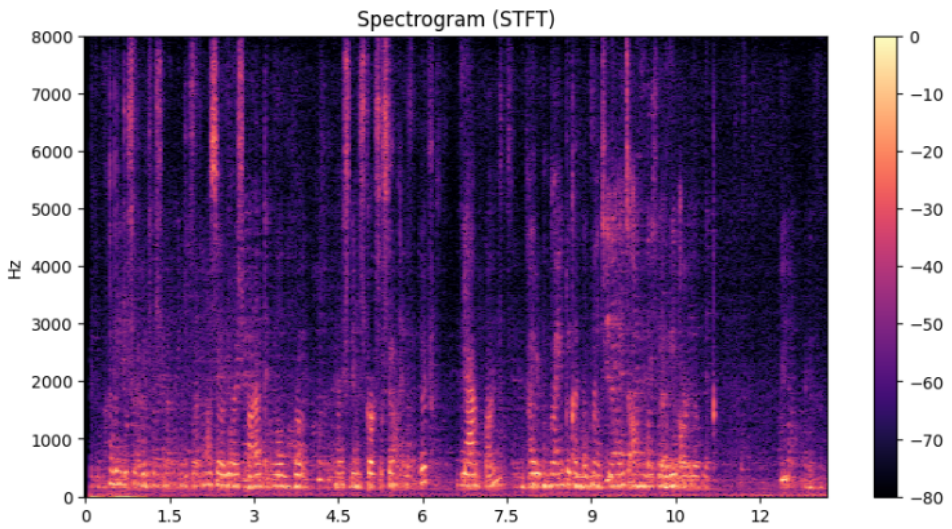


Figure 6: Spectrogram using STFT

**Observation:** Horizontal bands indicate formants, and vertical structures indicate speech transitions.

## 5.7 Narrowband Spectrogram

Narrowband

```
D_narrow = librosa.stft(y, n_fft=2048)
S_narrow = librosa.amplitude_to_db(np.abs(D_narrow))

plt.figure(figsize=(10,5))
librosa.display.specshow(S_narrow, sr=sr, x_axis='time', y_axis='hz')
plt.title("Narrowband Spectrogram")
plt.colorbar()
plt.show()
```

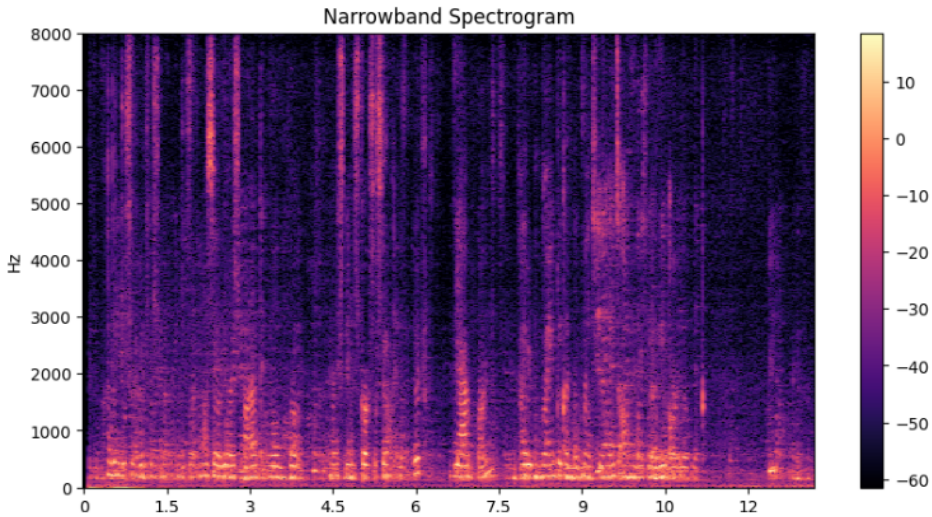


Figure 7: Narrowband Spectrogram

**Observation:** Harmonics are clearly visible due to better frequency resolution.

## 5.8 Wideband Spectrogram

wideband

```
D_wide = librosa.stft(y, n_fft=256)
S_wide = librosa.amplitude_to_db(np.abs(D_wide))

plt.figure(figsize=(10,5))
librosa.display.specshow(S_wide, sr=sr, x_axis='time', y_axis='hz')
plt.title("Wideband Spectrogram")
plt.colorbar()
plt.show()
```

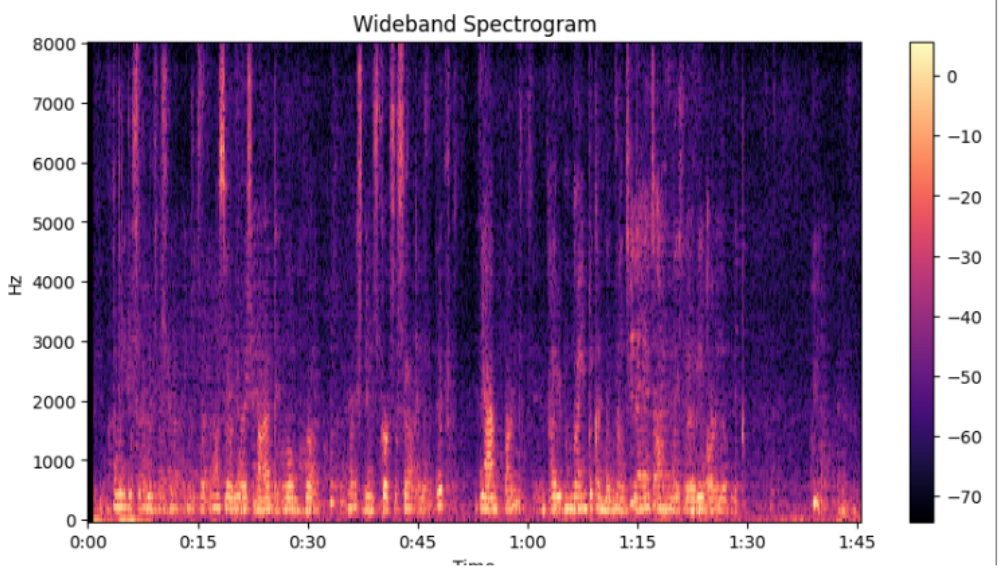


Figure 8: Wideband Spectrogram

**Observation:** Formants are more visible due to better time resolution.

# 6 Analysis

- FFT provides global frequency information but lacks time resolution.
- STFT provides both time and frequency information.
- Pitch contour shows variation of fundamental frequency over time.
- Harmonics appear as multiple peaks in FFT.
- Formants appear as horizontal bands in spectrogram.
- Narrowband spectrogram highlights harmonics clearly.
- Wideband spectrogram highlights formants clearly.

# 7 Inference

- Pitch, harmonics, and formants are distinguishable using FFT and STFT.
- FFT is suitable for frequency analysis, while STFT is suitable for time-frequency analysis.
- Spectrograms provide a comprehensive understanding of speech characteristics.
- The choice of window size affects resolution (time vs frequency trade-off).

# 8 Conclusion

In this experiment, pitch, harmonics, and formants were successfully extracted using FFT and STFT. Spectrogram analysis provided insight into speech characteristics, and the difference between narrowband and wideband representations was clearly observed.

# 9 Learning Outcomes

- Learned how to process speech signals using Python
- Understood FFT and STFT concepts practically
- Gained insight into pitch, harmonics, and formants
- Learned how spectrograms represent speech signals
- Understood the trade-off between time and frequency resolution